



Sleek book reader

Plastic Logic has a nice looking prototype of its soon to be released Que ebook reader with a capacitive screen

Thus simply designing with the end user in mind. Ergonomics must also closely follow aesthetics. For instance you can't make a digital camera that looks fantastic but is just too difficult to hold. However depending on function, following aesthetics, ergonomics is the first one to take a back seat in certain cases. Cars designed to achieve high speeds may not be very ergonomic i.e. comfortable to use, but they serve their purpose.

Much of the products of today are electronics. Although interface design does not strictly come under the purview of ergonomics it goes a long way into the usability of a product. Interface design has to work in tandem with the limitations or capability of the underlying hardware. "With the advancements in



One of the early prototypes of a typewriter

computing, processing is moving towards multi-core and multi-chip configurations. We constantly work with hardware manufacturers and do customise our software depending on factors such as the instruction set of the processor. But overall, we work towards the best for our user", said Ajay Pande, VP Engineering, Adobe India.

Prototyping and testing

In Donald Norman's legendary book titled "The Design of everyday things" the author quotes an interesting story about the evolution of the modern-day typewriter. The inventors conceptualised, designed and redesigned their creation till about thirty experimental devices or *prototypes* were made. Each one was slightly different and potentially incrementally better than the other. These products were then sent to stenographers and people from other walks of life. A particular one amongst these, a Mr. James O. Clephane, drove the instruments so hard that he ended up breaking them faster than the creator was able to supply them with more pieces (to wreck). Yet with the help of these prototypes, the



The device helps position a knife properly in front of the rocker knife as well as a tool to help hold food

creators were able to spot functional defects and the end result was a fool-proof machine.

Why is design important in today's world?

From a manufacturer's point of view "Design is the only differentiator when companies reach a common technological platform. It is that aspect of a product which first attracts a customer towards

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itself," says Sandip. It helps the end user as he gets access to a much better product. Himanshu Mishra, a Product Designer from CPDM, IISc, Bangalore who is currently associated with Innovation Labs - Tata Elxsi, echoes this sentiment. "Being faced with enormous choices for each of the commodities in today's world, one key differentiator that has evolved is

design, which causes the consumer's decision to tip in its favour. But more importantly, design is crucial not merely as an act of making commodities more appealing, but also to cater more as a problem solving approach to the issues that our society faces in this century. Thus, design becomes an all pervasive act required to strategically create or "design" products, services, systems, tools or any other manifestation that solve the omnipresent problems and addresses to needs and aspirations of the individuals, businesses, the society and the ecology," he says.

So as we've come to realise design can be put to use for the greater good of society too. "I think that in today's world consumers need something to excite them. but good design is life changing," said Hayley S. Rosen, Industrial Design student at Philadelphia University. "During my third year, we did a project with a group of occupational therapists and got an opportunity to work with clients that have various medical needs and couldn't do certain things. My partner Dan and I had three clients with three different needs. We were able to make prototypes and test them so we could make a final product. For example, we had a client that couldn't use a knife properly due to the lack of muscular development in his hands. We ended up taking an already existing product and adding a piece on to it so he could position a fork properly so he could cut his food. The real prize was when he used the device and was able to cut his food better than he could. It was very rewarding to see that his confidence level went through the roof because he could now do such a seemingly simple task without any problems".

So what are the challenges designers face?

The natural process of evolution of products is a slow and time consuming one. In the market conditions that designers operate in, there is rarely that liberty. The pressures include demands of time, forced innovation to come up with a new release of the product every year. Plus designers have to conform to the design philosophies of their organisation while retaining their individual mark. In the cutting edge world of technology designers also have to put more juice into a gadget while working within the pressures of technical limitations. Design being an activity which has a myriad range of stakeholders from people who manufacture it, market it, buy it, take care of after sales service to people who recycle